

# Ayyaz Akin

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## /\*\*\*\*\* Summary \*\*\*\*\*/

Infrastructure · Security · Routing · Automation · DevEx · Observability · Incident Management

Hands-on software engineer with 10+ years of experience designing, building, and scaling robust distributed systems, developer platforms, and high-impact software architectures. Proven ability to drive cross-functional initiatives, mentor engineering teams, and deliver highly available, secure, and performant systems. Adept at bridging technical vision with business goals through strong communication and strategic leadership.

## /\*\*\*\*\* Education \*\*\*\*\*/

[09/23-05/25] MS Cybersecurity@New York University:~/New York/NY ./gpa 3.90

- > Improved XMR blockchain mining throughput by over 1000% through advanced operating system tuning, process scheduling optimization, clustering, and efficient hardware utilization.
- > Implemented secure multi-cloud hybrid infrastructure using VPN tunneling, network segmentation, two-factor authentication, and comprehensive service/host monitoring.
- > Discovered vulnerable open endpoints within the NYU domain through active reconnaissance, followed by penetration testing reports detailing risks and remediation strategies.
- > Hosted a global Capture the Flag competition with over 1,800 teams and 4,500 players, building hybrid infrastructure and creating multiple challenges across web and crypto.
- > Researched wireless network attacks, including deauthentication, spoofing, and rogue access points (APs), to understand vulnerabilities and assess mitigation strategies.

[09/14-05/17] MS Information Technology@Virginia Tech:~/Blacksburg/VA ./gpa 3.97

- > Developed predictive models to forecast customer tipping behavior by researching and integrating public NYC taxi trip data with meteorological weather conditions.
- > Applied graph theory and data mining techniques to customer-escort networks to detect communities, optimize recommendations, and track potential viral transmission pathways.

[09/11-05/14] BS Information Technology@George Mason:~/Fairfax/VA ./gpa 3.32

- > Improved RSS feed data delivery by building a responsive web and mobile app that geo-populates and visualizes missing children data on an interactive map.

## /\*\*\*\*\* Profession \*\*\*\*\*/

[09/21-Today] Lead Software Engineer (DevOps)@The Walt Disney Co.:~/New York/NY

[11/19-09/21] Sr. Software Engineer (DevOps)@The Walt Disney Co.:~/New York/NY

- > Served as technology lead, collaborating across product, platform, and engineering organizations to deliver cohesive web infrastructure for millions of customers.
- > Designed and deployed a hybrid staging environment with support for ambiguous multi-origin routing leveraging DNS split-view architecture to simplify multi-region failover.
- > Built a scalable static web proxy to serve internal documentation and apps with minimal overhead, which is gated by SSO authentication to allow access for the entire company.
- > Increased web throughput by 70% by optimizing caching layers, resolving bandwidth limitations, and eliminating performance bottlenecks across critical infrastructure.
- > Implemented a flexible polymorphic logging system across web applications and seamlessly integrated it with centralized observability tools to enhance traceability and monitoring.
- > Reduced pipeline runtimes by 20+ minutes by implementing a remote cache solution and layering additional caching mechanisms to accelerate stages across CI/CD workflows.
- > Configured IP-based access control using edge cloudlets and implemented custom redirect rules to enforce secure access, and support dynamic traffic handling across environments.
- > Streamlined subdomain structure to enable secure, scalable, and PCI-compliant integration with external systems, reducing complexity and improving cross-origin communication.
- > Architected a self-service routing portal to empower engineering teams with dynamic route management capabilities, enabling quick, safe testing and increased operational autonomy.
- > Supported developers with hands-on debugging, architectural and implementation guidance, and aligning technical solutions with evolving product goals and cross-team requirements.

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## /\*\*\*\*\* Profession \*\*\*\*\*/

[01/18-10/19] Sr. Software Engineer (Full-Stack)@Capital One:~/McLean/VA

[11/16-01/18] Software Engineer (Full-Stack)@Capital One:~/McLean/VA

- > Improved first-time user experience by developing interactive, accessible navigation UI components using modern frontend frameworks, enhancing usability and user onboarding.
- > Presented technical topics at internal professional networking events to foster knowledge sharing, encourage cross-team collaboration, and support a culture of continuous learning.
- > Upgraded cloud application architecture to support secure, scalable, and compliant CI/CD pipelines, aligning with enterprise standards, and reliable software delivery.
- > Enhanced the marketing decision engine by integrating real-time big data streaming solutions, enabling faster insights, improved user targeting, and responsive campaigns.
- > Made significant contributions to Capital One's enterprise Node.js development framework, driving consistency, reusability, and improved developer experience across internal apps.
- > Actively participated in multiple inner-source projects while mentoring junior developers, fostering collaboration, and promoting engineering quality and best practices.
- > Advanced developer CI/CD toolkits to streamline and automate serverless application delivery, testing workflows, and environment provisioning for reliable deployments.
- > Reduced monthly cloud costs by ~\$2,000 by modernizing legacy web applications using scalable, event-driven serverless technologies and eliminating unused infrastructure.

[01/16-11/16] Software Engineer II (Full-Stack)@Booz Allen Hamilton:~/McLean/VA

[08/14-01/16] Software Engineer I (Full-Stack)@Booz Allen Hamilton:~/McLean/VA

- > Actively supported rapid prototyping, requirements gathering, stakeholder presentations, and the implementation of key product features from concept to delivery.
- > Operated as a full-stack developer in a test-driven development environment, building scalable intelligence platforms that supported real-time data analysis and reporting.
- > Designed and implemented an automation tool for efficient data interrogation, integrity validation, and scheduled ingestion updates across large data sets and sources.
- > Overhauled an enterprise software system's dependency mapping and rebuilt its comprehensive testing suite to improve maintainability, and deployment confidence.
- > Developed internal tools that enabled analysts to access, filter, and visualize near real-time threat intelligence reports, improving response times and decision-making.
- > Revamped a legacy security platform's API by addressing critical vulnerabilities, implementing modern authentication, and enabling robust end-to-end encryption.
- > Integrated advanced tracking tools across multiple web and mobile applications to generate unified insights and surface actionable predictive business intelligence.
- > Reduced memory complexity from  $O(n^2)$  to  $O(1)$  by engineering a highly efficient, scalable streaming-based re-indexing solution for large data pipelines, reducing time to deploy.

## /\*\*\*\*\* Technologies \*\*\*\*\*/

Adobe....: Photoshop, Lightroom, Illustrator, Fresco

Client....: HTML5, CSS3, JavaScript, TypeScript, Next

Server....: Node, Bash, Groovy, Java, Python, Ruby, PHP

Database.: Auora, Dynamo, Redis, Mongo, Neo4j, Keyspaces

Analytics: Kinesis, MQ, EMR, Redshift, Athena, Data Lakes

AI.....: SageMaker, Scikit-learn, Network-X, TensorFlow

OS.....: Linux, MacOS, Windows, IOS, Android

Cloud.....: AWS, Azure, GCP, On-Premise, Hybrid

Security..: IAM, SCP, KMS, JWT, VPN, WAF, IR, SSO

Networking: Route53, RT, VPC, CIDR, ACL, SG, BGP, DNS

Monitoring: SNS, PagerDuty, Datadog, ELK, RUM, APM, XRay

Automation: CloudFormation, Jenkins, Docker, Hashicorp

## /\*\*\*\*\* Certifications \*\*\*\*\*/

AWS: Associate (Solution Architect, Developer, SysOps)  
Professional (Solution Architect, DevOps Engineer)  
Specialty (Security, Advanced Networking)

Scrum..: Certified Scrum (Master, Developer, Product)  
CompTIA: Infrastructure (Network+, Linux+, Cloud+)  
Cybersec. (Security+, CySA+, PenTest+, CASP+)

## /\*\*\*\*\* Github \*\*\*\*\*/

disneystreaming/automated-cloud-advisor

neetcode-gh/leetcode

osirislab/CSAW-CTF-2023-Finals

aakhtar3/aakhtar3.github.io

aakhtar3/javascript-projects

aakhtar3/school-projects